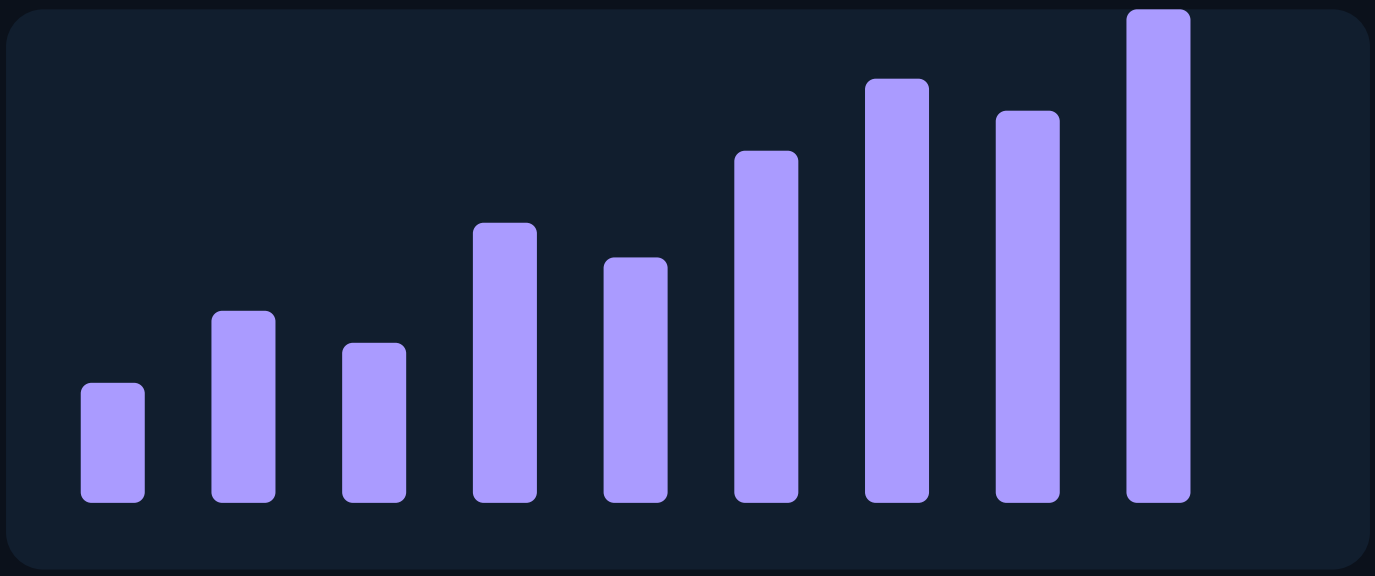
PRACTICAL KNOWLEDGE / DARK EDITION

Microsoft Fabric Foundations

Design a small analytics platform from ingestion to a trusted semantic model, with practical checkpoints at every boundary.

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DATTA SABLE / PROFESSIONAL FIELD GUIDES

THE READER BRIEF

A guide you can put to work

Fabric brings several analytics workloads into one platform, but integration does not remove the need for architecture. A useful solution still needs a defined source contract, a storage and transformation design, a governed semantic model and an operating process.

This guide follows an invented distributor, Northstar Supply, that receives daily order-line extracts and wants a reliable sales report. The goal is not to activate every Fabric feature. It is to explain each boundary clearly enough that a second engineer can build, verify and operate the design.

The guide is independently authored and is not an official Microsoft course or certification resource. Interface names and platform behaviour should be checked against the linked Microsoft Learn documentation for your tenant and edition. No capacity price or performance outcome is promised.

What you need

Basic SQL and data-model familiarity. A Fabric-enabled practice environment is needed for platform exercises; licensing, capacity, tenant settings and feature availability depend on your organisation. The architecture and paper exercises can be completed without provisioning resources.

Edition and scope

Original educational examples by Datta Sable. First edition, 28 October 2026. This purchase contains this PDF only. Code, tables and exercises are printed inside; no separate datasets, software licences or editable files are bundled. Product interfaces and availability can change. Check the linked official documentation when applying a platform-specific step.

Personal learning licence

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FIELD GUIDE / 01

Start with the business and service contract

Northstar Supply wants yesterday's paid merchandise revenue by product and region by 09:00 each business day. Its source system exports order lines after midnight. Finance needs a report that can explain adjustments and distinguish late input from zero activity.

Write the acceptance criteria

Freshness: the reporting model contains the agreed source batch through the prior business day.
Completeness: source and accepted-plus-rejected row counts reconcile.
Correctness: the same paid-revenue definition is used in the curated table and semantic measure.
Access: regional readers see only the region they are authorised to view.
Recovery: a failed batch can be replayed without duplicating sales.

Those criteria determine architecture more directly than a list of product features. A five-minute ingestion interval does not help if the upstream extract is produced once daily. A beautiful report does not compensate for a missing batch.

Separate three identities

Source business key: order_id plus line_no.
Ingestion identity: batch_id plus source filename.
Reporting identity: the fact row and its dimension keys.

Keeping them distinct makes retries, corrections and lineage easier to explain. A source file name is not a reliable substitute for a business key.

Exercise and answer

The source file is absent at 08:30. Should revenue be zero? No. Mark freshness as failed or incomplete, retain the last valid data with its timestamp and investigate source delivery. Missing evidence is not evidence of no sales.

FIELD GUIDE / 02

Map the platform to the workload

Fabric includes integrated experiences for ingestion, engineering, warehousing, real-time work, data science and reporting over shared platform services. OneLake provides a common logical data-lake foundation. The important design question is which experience fits your team's data and operating model.

A restrained reference architecture

Source extract -> ingestion process -> raw landing data -> validated, conformed tables -> semantic model -> report.

For Northstar, batch orchestration can coordinate ingestion and transformations. A lakehouse can support file-oriented landing and engineering workflows. A warehouse can suit SQL-oriented dimensional serving. A semantic model owns business measures and reporting relationships. You may use fewer components if one experience meets the requirement cleanly.

Lakehouse or warehouse?

Compare the team's skills, required transaction and query patterns, data formats, transformation tools and governance needs. Do not duplicate the same curated data across experiences without a reason such as a specific serving requirement. Every extra copy adds a refresh and reconciliation responsibility.

Avoid feature-driven architecture

Real-time processing is justified when low-latency events change an operational decision. A daily extract does not become real time because it passes through a streaming-labelled component. Likewise, notebooks are useful for code-driven work, but SQL transformations may be easier for a SQL-focused support team.

Exercise and answer

List the owner and verification evidence for each arrow in the reference architecture. Any arrow without an owner is an operational gap, even when the platform makes the connection easy to create.

FIELD GUIDE / 03

Organise workspaces and environments

A workspace is a collaboration and management boundary, not merely a folder. Plan development, validation and production responsibilities before creating items. Separate experiments from production delivery so a practice refresh cannot overwrite a trusted reporting table.

A simple ownership map

Platform owner: capacity and tenant-level controls.
Data owner: source meaning, permitted use and quality decisions.
Engineering owner: ingestion, transformations and recovery.
Model owner: relationships, measures and report semantics.
Support owner: alert triage and incident communication.

One person can hold several roles, but each responsibility needs an explicit name. Access should follow the task being performed. A report consumer does not automatically need engineering edit rights.

Name for operations

Use names that reveal purpose and environment, such as sales-dev and sales-prod, and item names such as ingest_order_lines or dim_product. Keep secrets and connection details outside notebooks and code. Document how connections and parameters differ between environments.

Promotion is a testable event

A release should identify the source version, changed items, schema compatibility, target environment and rollback approach. Verify that a promoted process points to the intended production source and destination. A successful item deployment does not prove the connection parameters or access policies are correct.

Exercise and answer

An analyst asks for workspace administrator access to view one report. Identify the smallest supported permission path that satisfies the viewing requirement, then verify it with an appropriate test identity. Do not use broad edit rights to compensate for unclear sharing design.

FIELD GUIDE / 04

Land raw data with provenance

A raw layer is evidence of what arrived. Preserve the received values and useful provenance rather than silently replacing files with cleaned output. That evidence is essential when a late correction changes a month already reported.

Minimum batch metadata

Record batch_id, source_system, source_filename, arrival_time_utc, source_period, byte_size, checksum where practical and the ingestion outcome. Record row counts after parsing. Keep the original file or an agreed immutable representation according to the organisation's retention policy.

Define replay behaviour

If the same batch arrives twice with the same checksum, the process should recognise it rather than append duplicate facts. If the same business period arrives with a different checksum, classify it as a correction or conflict under a documented rule. A filename alone is not enough to distinguish those states.

Validate structure before business rules

Check encoding, file type, required columns and parseable values. A structural failure can invalidate the entire batch. A row-level business defect may instead produce a quarantine row if the contract permits partial acceptance. Whatever the policy, accepted plus rejected row counts must equal parsed source rows.

Exercise and answer

Yesterday's file is delivered again under a new filename. If ingestion identity uses filename only, the batch may be duplicated. Use both provenance and stable source keys, then apply the agreed replay rule. Verify the final fact count and revenue remain unchanged.

FIELD GUIDE / 05

Build conformed tables with explicit quality gates

A common medallion-style pattern separates raw evidence, validated/conformed data and business-ready serving tables. The names bronze, silver and gold are useful shorthand, but each layer needs a real contract. Renaming a folder does not create quality.

Northstar's conformance rules

Order_id and line_no identify a source line.
Quantity must be positive for this sales-only source.
Unit_price and discount must be valid decimal values.
Paid status determines inclusion in the revenue metric.
Product and region keys must match an approved dimension or enter a visible unknown bucket.
Dates use the agreed reporting calendar and time zone.

Separate defects from transformations

Trimming whitespace is a transformation. A missing product key is an exception requiring a policy. A duplicated business key is a conflict requiring a deterministic winner or rejection. Do not place all three under an opaque "clean data" step.

Track quality as data

Store check_name, batch_id, observed_count, expected_rule and outcome. Keep row-level exception references where permitted. A threshold should have a reason: a 1% rejection rate may be unacceptable when the rejected records contain the largest transactions.

Exercise and answer

A batch has 1,000 parsed rows, 997 accepted and 3 rejected. Row counts reconcile, but accepted revenue is 30% below the source total. Investigate the rejected amounts and source definition. A low rejection percentage is not sufficient evidence that the monetary report is usable.

FIELD GUIDE / 06

Model facts and dimensions for reporting

A star schema separates measurable events from descriptive attributes. For Northstar, FactSales has one row per accepted order line. Dimensions describe Date, Product, Customer and Region. Declare the grain in the table documentation before choosing measures.

Facts should be additive where intended

Store quantity, line gross amount, line discount and paid net amount with clear types and definitions. Ratios such as margin percentage should generally be recomputed from their components at the requested aggregation level. Averaging row-level percentages can produce an incorrect total.

Dimension keys and unknown members

Use stable dimension keys and check uniqueness. If a source product cannot yet be mapped, an explicit Unknown member can preserve the fact while exposing the quality issue, if the business approves that policy. Hiding unmatched facts can make a report look cleaner while understating sales.

Historical attributes

A customer's current region may differ from the region at the sale date. Decide whether historical reports should restate under the current region or preserve the original classification. A slowly changing dimension design may be appropriate when the history itself is part of the reporting requirement.

Exercise and answer

You join a product dimension containing two rows with the same ProductKey to FactSales. Revenue doubles for that product. Fix dimension uniqueness or the historical-key join. SUM(DISTINCT Amount) is not a valid repair because two real lines can have identical amounts.

FIELD GUIDE / 07

Create a semantic model with a metric contract

A semantic model should centralise the meaning of a measure so each report does not invent its own variation. Keep relationships, filter direction and date behaviour understandable. Test a simple table visual before designing a dashboard.

```
Paid Revenue =
CALCULATE(
    SUM(FactSales[NetAmount]),
    FactSales[Status] = "paid"
)

Paid Orders =
CALCULATE(
    DISTINCTCOUNT(FactSales[OrderID]),
    FactSales[Status] = "paid"
)

Average Order Value =
DIVIDE([Paid Revenue], [Paid Orders])
```

These DAX examples assume the named columns exist and the relationship model is already valid. NetAmount is a line-level amount after discount, excluding tax and shipping. DISTINCTCOUNT is necessary because the fact grain is order line, not order.

Reconcile outside the report

Compare a semantic measure against a curated-table query for the same filters, cutoff and data version. Then test by date, region and product, including Unknown. A matching grand total can conceal wrong allocation across categories.

Storage and access choices

Evaluate the supported semantic-model storage mode for your data, capacity and access pattern. Direct Lake, import and DirectQuery have different operating considerations. Confirm the current official documentation rather than treating one mode as universally fastest or free of refresh responsibilities.

Exercise and answer

Two orders contain three and one lines respectively. COUNTROWS(FactSales) returns four; Paid Orders should return two. Put both values in a test table to make the grain distinction visible.

FIELD GUIDE / 08

Design security across every access path

Protecting a report is only one part of protecting an analytics platform. A user may reach data through a semantic model, SQL endpoint, notebook, workspace item or underlying storage path, depending on permissions and platform configuration. Review those paths explicitly.

Map identity to task

Readers consume approved reports.
Analysts build measures and reports within their scope.
Engineers operate pipelines and curated tables.
Administrators manage platform settings and permissions.

Use groups and documented role ownership where practical. Avoid sharing individual credentials or embedding secrets in notebook text. Keep connection authentication in the platform's supported credential mechanism.

Row-level rules need verification

If regional readers should see only their region, define the rule in the intended serving layer and test with representative identities. Verify totals, drillthrough, exports and the effect of membership in more privileged roles. A screenshot from an administrator account is not evidence of a reader's effective access.

Classification and retention

Identify personal or confidential fields before landing them. Retain only what the reporting purpose requires, and document deletion or retention responsibilities across copies. A raw evidence layer still needs appropriate access and retention controls.

Exercise and answer

A report is region-filtered, but the same reader can query the entire underlying table through another granted path. The requirement is not satisfied. Reduce or redesign that alternate access and verify the complete path again, rather than relying only on the report filter.

FIELD GUIDE / 09

Operate for freshness, recovery and capacity

A successful pipeline run proves that configured steps finished. It does not prove the intended source period arrived or the report reflects it. Observe technical execution and business freshness separately.

Minimum operating evidence

Last expected and received batch.
Rows parsed, accepted and rejected.
Revenue reconciliation result.
Transformation and model-update completion.
Data version visible to readers.
Duration, failures and retry count.

An alert should identify the failed boundary and the next owner. "Pipeline failed" without a source, batch or stage often leads to slow diagnosis. Avoid retrying a non-idempotent append blindly; replay safety must be designed before automatic retries.

Capacity is a workload constraint

Coordinate heavy ingestion, transformation and reporting workloads. Measure duration and resource behaviour using the platform's available monitoring tools. Representative data volumes and concurrency matter more than a small demonstration query. Licensing and capacity choices should be evaluated against the current tenant requirements and official guidance.

Recovery rehearsal

Choose a completed batch and replay it in a practice environment. Row count and revenue should remain unchanged. Then simulate a transformation failure after landing. Recovery should resume or rerun from a documented point without losing the original evidence.

Exercise and answer

The pipeline finishes at 08:00 but the report still shows data through two days ago. Investigate source period, curated-table version and semantic serving state. Do not mark the freshness target met solely because the pipeline status is green.

FIELD GUIDE / 10

Capstone: the Northstar release review

Produce a two-page architecture note and a small demonstrator or paper walkthrough. The note should let another engineer follow one order line from source file to report measure, including its security and recovery rules.

Deliverables

Architecture: name each component and its purpose.
Contracts: define source grain, fact grain, metric and freshness.
Quality: show row-count and amount reconciliation.
Security: document reader and operator access paths.
Operations: explain replay, alerting and rollback.
Evidence: record the batch and data version tested.

A review scenario

Batch B17 contains 100 source lines totalling 50,000 in paid net amount. Two lines with missing product mappings total 800. The business-approved policy retains them under Unknown. The curated fact should still contain 100 lines and 50,000. A product view should expose Unknown = 800, not silently drop it.

If the same batch is replayed, those figures must remain unchanged. If a corrected batch arrives with a changed checksum, the correction rule must determine whether it replaces, versions or rejects the previous data. That decision belongs in the source contract, not in an emergency notebook edit.

Acceptance answer

The design is ready for a controlled release when the same metric reconciles across layers, reader permissions behave as intended, freshness is visible and recovery is repeatable. A diagram alone cannot establish those properties. Keep the evidence next to the release version so future changes can be assessed against it.

KEEP LEARNING

Reference desk

Official references checked while preparing this edition. Use these pages for syntax, feature availability and the latest product details. The exercises and business scenarios in this book are original.

Microsoft Learn: Fabric overview

https://learn.microsoft.com/en-us/fabric/fundamentals/microsoft-fabric-overview

Microsoft Learn: OneLake overview

https://learn.microsoft.com/en-us/fabric/onelake/onelake-overview

Microsoft Learn: lakehouse overview

https://learn.microsoft.com/en-us/fabric/data-engineering/lakehouse-overview

Microsoft Learn: star schema guidance

https://learn.microsoft.com/en-us/power-bi/guidance/star-schema

Microsoft Learn: Direct Lake overview

https://learn.microsoft.com/en-us/fabric/fundamentals/direct-lake-overview

Make the next step observable

Write the Northstar metric and batch contracts before creating items. Build the smallest path that can demonstrate reconciliation, permission checks and a safe replay. Expand the platform only when a measured requirement justifies the next component.

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